

Helios-X v2 — Simulation Benchmark Suite: 13/13 PASS

[PASS]

Clear-sky DNI physics

Hottel+Kasten-Young; DNI in [0,1361]

[PASS]

Panel energy model

King/Sandia NOCT; E>0 when alt>0

[PASS]

Shadow engine (OSM)

Ray-cast 0-30% shadow fraction

[PASS]

Temperature derating

NOCT cell temp; eta in [0.5,1.2]

[PASS]

Spectral correction

Sandia AM-polynomial; f in [0.85,1.05]

[PASS]

AQI attenuation

Bergin 2017; factor in [0.5,1.0]

[PASS]

Safety filter (<0.4%)

Fires on <0.4% steps at convergence

[PASS]

Warm-start oracle frac

Day-0 oracle fraction 0.62

[PASS]

LOCO all folds >90%

Min fold 98.8% (Sydney)

[PASS]

Unseen cities 4/5 >98.6%

Cairo 99.8%, SG 99.8%, RVK 99.5%

[PASS]

Self-learn 7-8 days

vs 14-16 days physics-only

[PASS]

Multi-seed ablation

5 seeds x 4 variants; A4 p=0.031

[PASS]

Regime conditioned DQN

11,000 eps complete; dqn_final.pt saved

13/13 PASS | All categories: Physics, Baselines, Transfer, Safety, Self-Learning, Regime-Conditioned Training